CSE 225 Data Structures
Week #8

A
Quiz #3

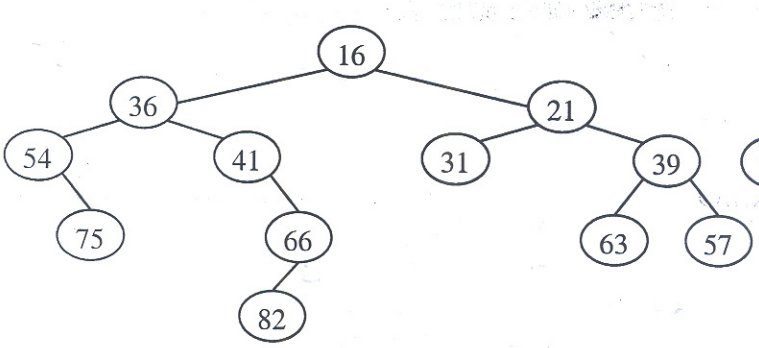
Borahan Tümer, Ph.D.
Fall 2006

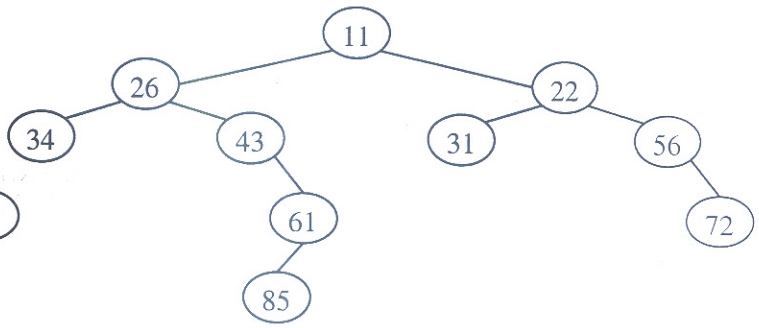
Q1.

Two heaps are given below.

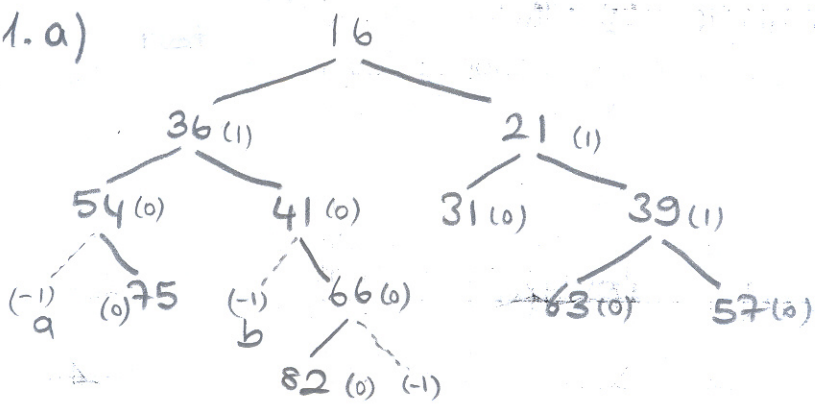
a) Are they leftist heaps?

b) Merge them ***recursively*** in ***O(lg n)*** time where ***n*** is the number of elements in both heaps! Make any (or both if necessary) in case they are not LHs!





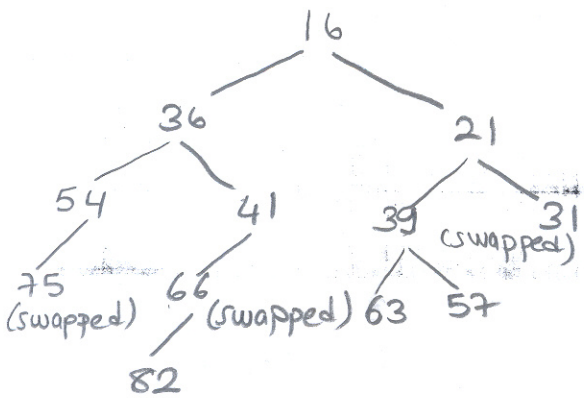
1.a)



Not leftist : 1) Npl (31) < Npl (39)
2) Npl (a) < Npl (75)
3) Npl (b) < Npl (66)

* Second one is not leftist, either. The solution is on the (B) group pages.

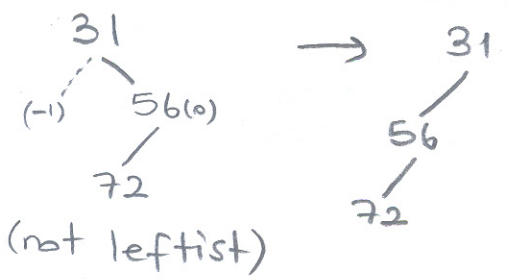
1.b) First, make the heaps leftist.



* Solution for the second one can be found in (B) group pages.

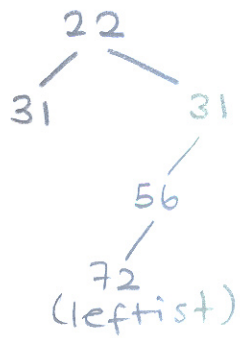
1.b.) cont. Comparison table: 16 - 11 (A)
16 - 22
21 - 22
22 - 31
31 - 56

1)



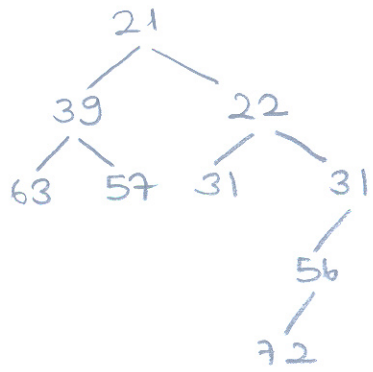
(not leftist)

2)



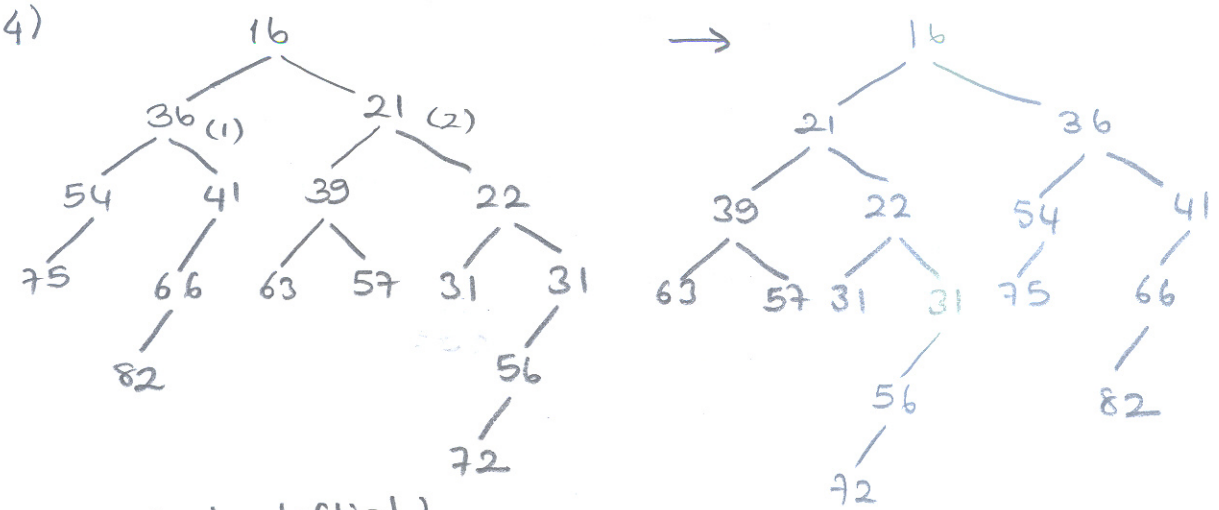
(leftist)

3)



(leftist)

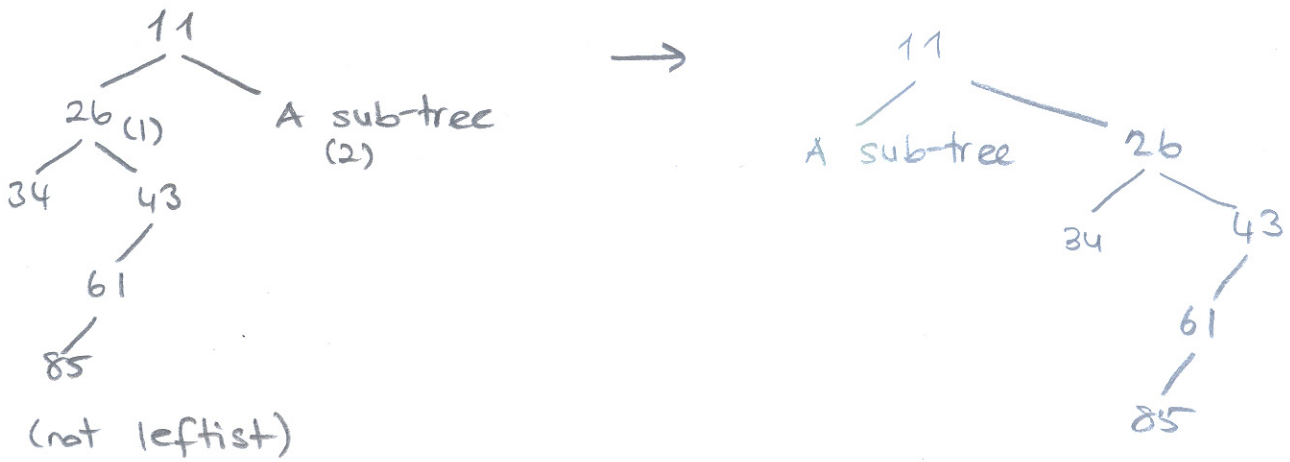
4)



(not leftist)

(A sub-tree)

5)



(not leftist)

(result)

Q2.

Consider a binomial heap of 4223 nodes.

a) How many binomial trees does it have?
b) What are these trees?
c) How many nodes does the "highest" (or "deepest") binomial tree have in its ***most crowded depth level***? (most crowded: en kalabalık)

2.a) $4223 = 2^{12} + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$

So, it has 8 binomial trees.

2.b) $B_0, B_1, B_2, B_3, B_4, B_5, B_6, B_{12}$

2.c)
- highest binomial tree is B_{12}
- most crowded depth level is 6th level
- it has:

$$\binom{k}{d} = \binom{12}{6} = \frac{12!}{6!\,6!} = 924 \text{ nodes}$$